

Question description

You are given a list consisting of n integers. Your task is to remove elements from the list at given positions, and report the removed elements.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq p_i \leq n - i + 1$

Input

The first input line has an integer n : the initial size of the list. During the process, the elements are numbered $1, 2, \dots, k$ where k is the current size of the list.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the list.

The last line has n integers $p_1, p_2, \dots, p_n$: the positions of the elements to be removed.

Output

Print the elements in the order they are removed.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
2 6 1 4 2
3 1 3 1 1

EXPECTED OUTPUT

1 2 2 6 4

Test Case 2

INPUT (STDIN)

6
7 2 6 1 4 2
11 3 1 3 1 1

EXPECTED OUTPUT

2 6 7 4 2 1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void build(int k,int
l,int r)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

330

Test Case 3

NLOC

37

Tree110.c +

```
1 #include <stdio.h>
2
3 #define MAXN 200005
4
5 int arr[MAXN];
6 int bit[MAXN];
7 int n;
8
9 void update(int i, int x) {
10     while (i <= n) {
11         bit[i] += x;
12         i += i & (-i);
13     }
14 }
15
16 int find_kth(int k) {
17     int idx = 0;
18     int step = 1;
19     while (step * 2 <= n) step *= 2;
20     for (; step >= 1; step /= 2) {
21         int nx = idx + step;
22         if (nx <= n && bit[nx] < k) {
23             k -= bit[nx];
24         }
25     }
```

```
26     }
27     return idx + 1;
28 }
29
30 int main() {
31     int i;
32     if (scanf("%d", &n) != 1) return 0;
33     for (i = 1; i <= n; i++) scanf("%d", &arr[i]);
34     for (i = 1; i <= n; i++) update(i, 1);
35     for (i = 0; i < n; i++) {
```

```
36     int p;  
37     scanf("%d", &p);  
38     int idx = find_kth(p);  
39     printf("%d ", arr[idx]);  
40     update(idx, -1);  
41 }  
42 printf("\n");  
43 return 0;  
44 }
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

5

2 6 1 4 2

3 1 3 1 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree110.exe

1 2 2 6 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

6

7 2 6 1 4 2

11 3 1 3 1 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Tree110.exe

0 6 7 4 2 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>